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| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 12/22/2016 |
| **PI Name:** | Laughlin, Mitzi Ph.D. | | |
| **Project Title:** | Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Biomedical countermeasures | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Web Page:** | | | |
| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77204-6015 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2010 Crew Health NNJ10ZSA003N |
| **Start Date:** | 02/23/2016 | **End Date:** | 08/24/2020 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Norsk, Peter | **Contact Phone:** | |
| **Contact Email:** | Peter.norsk@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: End date changed to 8/24/2020 per HHC element/JSC and NSSC information (Ed., 12/10/20) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Clarke, Mark SF Ph.D. ( University of Houston )<br>Layne, Charles S Ph.D. ( University of Houston ) | | |
| **Grant/Contract No.:** | NNX16AF63G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Use the Muscle Atrophy Research and Exercise System (MARES) device to quantify the time course of changes in muscle strength and performance of crewmember volunteers during six-month International Space Station (ISS) missions. Establish baseline strength and performance capabilities of each participating crewmember from two pre-flight test sessions, track changes from that baseline during three in-flight test sessions, and track recovery to baseline levels during five post-flight test sessions.<br>The project is currently ongoing and being coordinated by the European Space Agency (ESA) for flight. Several meetings have taken place to validate the ground and flight data collection procedures and another is scheduled in August at the European Astronaut Centre in Cologne.<br><br>Continuation of Modulation of Muscle Function by Lower Limb Loading during Space Flight, grant NNX12AF04G. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The proposed investigation seeks to systematically examine the relationships among muscle strength and neuro-motor control performance. By tracking these relationships over time during flight we will be able to identify any adaptive changes in integrated motor performance. If successful, this investigation should provide significant benefits to the Human Research Program by quantifying the aspects of the Program Requirements Document (PRD) Risk of Impaired Performance Due to Reduced Muscle Mass, Strength and Endurance embodied in IRP (Integrated Research Plan) Gap M24 (What is the time course of changes in muscle protein turnover, muscle mass and muscle function during long-duration spaceflight?) and M2 (What is the current status of in-flight/post-flight exercise performance capability?). Additionally, this investigation will provide valuable information to clinicians regarding muscle atrophy due to injury or disease on Earth. |
| **Task Progress:** | The grant NNX16AF63G, "Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES," began in February 2016 and is approaching the one-year anniversary. The study is designed to examine declines in muscular strength, function, and neuromotor control during long-duration spaceflight using the MARES device in the Columbus module on the ISS.<br>Extensive progress has been made in the first year with the international protocol and agreements as well as study start with three consented subjects. The SarcoLab-3 team which includes this project, ESA SarcoLab project, and Russian scientists from IBMP (Instiute of Biomedical Problems) met several times in the spring to develop a common protocol that met each of our research objectives. These meetings were facilitated by CADMOS and the Senar engineers as they were developing the software and assisting in trouble-shooting MARES. There were no issues with MARES itself, rather we as scientists had difficulty getting MARES to perform the testing in a manner consistent with our laboratory methods. The timing of force application, range of motion stops, and ramping the force applied to the subject gradually were the main issues encountered. All these issues were overcome and programmed into the final protocol for SarcoLab-3. There is a dedicated ground protocol and another flight protocol. All tests are conducted in the ground protocol but only a subset of tests are included in the flight protocol to be time efficient.<br><br>Since this is an international project, ESA has taken the lead on preparing MOUs (memorandum of understanding) for the use of data and publications. CADMOS has been tasked with archiving all study data and will distribute all data to all scientists. A signed MOU details which scientists are responsible for analysis of specific tests and data. Once analysis is complete, scientists are responsible to send the results to everyone along with the R code used in the analysis or in the case of images the image processing software. The first publication is planned to be a joint publication with all three research teams involved and a pooling of data to publish in a very high impact journal.<br><br>Currently there are 3 crew members who have consented for this study. The first subject has completed a chair adjustment session, all pre-flight training, two pre-flight BDC sessions, and the FD7-10 and FD45(+/-14) sessions. The processing of data for this single subject is ongoing and the data quality is excellent from the ISS. The second subject has completed a chair adjustment session and is in the process of completing the training with BDC sessions expected to be scheduled in the spring. The third subject has just consented and has not started study procedures yet. This subject will need to complete all study procedures in the spring prior to launch in May 2017. |
| **Bibliography Type:** | Description: (Last Updated: 12/10/2020) |
| **Abstracts for Journals and Proceedings** | Rittweger J, Laughlin MS, Kozlovskaya I. "Sarcolab-3: Myotendinous and neuromuscular adaptation to long-term spaceflight." Presented at the Joint ESA, NASA, JAXA and CSA Science Symposium, June 7-9, 2016.<br>Joint ESA, NASA, JAXA and CSA Science Symposium, June 7-9, 2016. , Jun-2016 |